Direct Visualization and Comparative Analysis of Let-7 and Lin-4 Transcriptional Dynamics in *C. Elegans* With MS2-MCP Signal Optimization

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Introduction

Developmental Timing

- Organisms undergo a series of **highly-regulated** developmental events [1]
- Timing is primarily controlled at the transcriptional level [1]
- **Transcription factors** bind to gene regulatory regions, generating regulatory RNAs to activate or repress gene expression, coordinating cell fate transitions [2]
- Deviations in developmental timing genes can **disrupt larval stage progression**

C. elegans

- Microscopic, transparent nematode with a fully mapped and **invariant cell lineage** [3]
- Short life cycle, about 2 weeks, enables rapid developmental studies
- Exhibits **four larval stages** (L1-L4) before adulthood [4]

Heterochronic MicroRNAs

- The Heterochronic Pathway controls developmental gene regulation in *C. elegans* [5]
- Lin-4 and let-7 regulate **stage-specific** developmental timing [6]
- **Lin-4** is expressed in an oscillatory pattern, with a single pulse of expression during each larval stage [7]
- **Let-7** expression is theorized to surge during L4, triggering **larval transition to adulthood** [8]

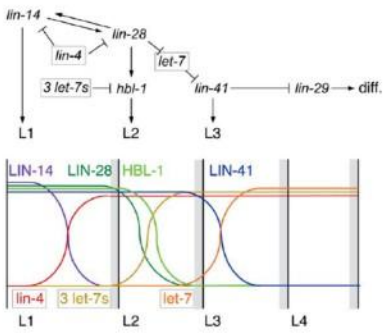


Figure 1. C. elegans miRNA Regulation in the Heterochronic Pathway. Graph generated by [9]

MS2-MCP System

- MS2 RNA stem-loops are inserted into endogenous genes via CRISPR [9]-[10]
- **MCP-GFP binds to MS2 hairpins during transcription**, forming nuclear fluorescent foci
- Nuclear background depends on MCP concentration and affinity

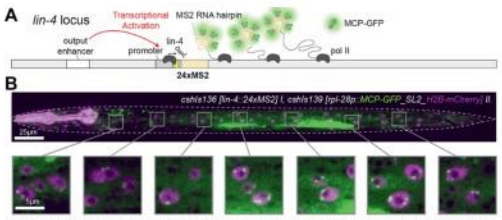


Figure 2. Visualization of transcription using the MS2-MCP system. Figure from previous Hammell Lab work [10]

Investigation Goals

- Visualize endogenous let-7 transcription for the first time
- Compare the transcriptional phase and amplitude between let-7 and lin-4
- Test balanced vs. unbalanced MCP expression
- Determine the effects of destabilized MCP (dMCP) compared to MCP

Methodology

1) Plasmid Construction

- Designed **let-7::24xMS2** plasmid in SnapGene
- Amplified homology arms, vector backbone, and MS2 cassette via PCR
- Gel-purified fragments and assembled using Gibson Assembly
- Verified plasmids using colony PCR, restriction digest, and sequencing

2) CRISPR and Injections

- Prepared **CRISPR injection** mix with Cas9, sgRNAs, plasmids, and rol-6 marker
- **Microinjected** mixes into N2 nematodes using micromanipulator
- Selected F1 rollers and genotyped to confirm successful MS2 insertion

3) Strain Generation

- Removed Self-Excising Cassette using heat shock induction
- **Generated** homozygous **MS2::let-7 and MS2::lin-4** reporter lines
- **Crossed reporters** with MCP-GFP or dMCP-GFP backgrounds
- Created **balanced and unbalanced MCP/dMCP strains**

4) Fluorescence Microscopy

- Immobilized L4 animals on agar pads with levamisole
- **Imaged** on Zeiss Axio Observer using 63x oil objective
- Collected DIC, GFP (MCP-GFP), and mCherry (H2B) channels

5) Quantitative Analysis

- **Processed image stacks** in Fiji (ImageJ, v2.9)
- Identified transcriptional foci via GFP/mCherry colocalization
- **Calculated signal-to-noise** ratio (SNR = $I_{spot}/I_{background}$)

6) Statistical Analysis

- Compared groups using **two-tailed Welch's t-tests**
- Calculated effect sizes using **Cohen's d**
- Graphed data using Prism v9 (GraphPad Software)

Results

- Live imaging revealed transient nuclear fluorescent MCP-GFP dots in developing larvae, corresponding to let-7 transcripts
- Nuclear MCP-GFP foci colocalized with H2B-mCherry, confirming real transcription activity
- Let-7 transcription primarily detected during the L4 stage, encompassing **L4.4** (n=114 total let-7 nuclei)

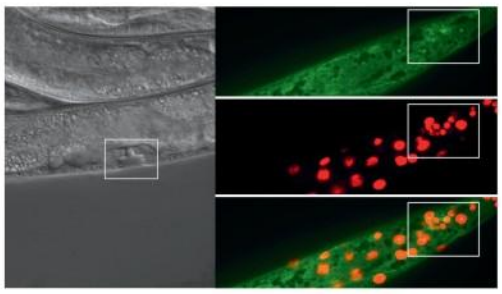


Figure 3. Visualization of endogenous let-7 transcription using the MS2-MCP system. Graph created by student research using FIJI (ImageJ), 2025.

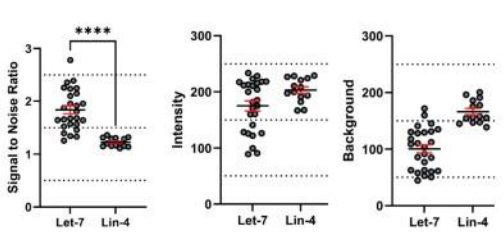


Figure 4. Quantification of nuclear MCP-GFP foci in strains expressing unbalanced MCP-GFP. Graph created by student research using GraphPad Prism v9, 2025. *Statistics: t(29.09) = 7.72, p < 0.001, d = 2.50*

- Let-7 MCP-GFP SNR = **1.837**
- **Let-7** exhibited **significantly higher SNR** than **lin-4** under unbalanced MCP-GFP expression
- Raw fluorescent intensities were similar, suggesting shared upstream transcriptional regulators

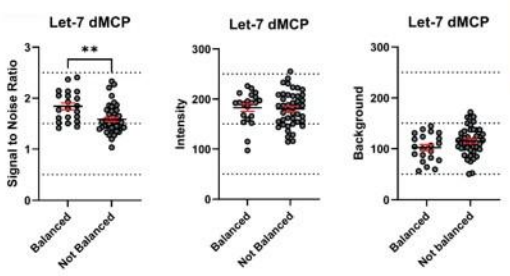


Figure 5. Effect of dMCP-GFP balancing on transcriptional signal quality of let-7 dMCP strains. Graph created by student research using GraphPad Prism v9, 2025. Statistics: t(37.80) = 3.285, p = 0.0022, d = 0.87

- **Balanced** dMCP-GFP strains showed **significantly higher** SNR than **unbalanced** dMCP strains
- Background intensity substantially decreased in balanced animals

Results (continued)

- **Balanced dMCP** showed **significantly greater SNR** relative to **balanced MCP**
- **Unbalanced dMCP** showed **significantly lower SNR** relative to **unbalanced MCP**

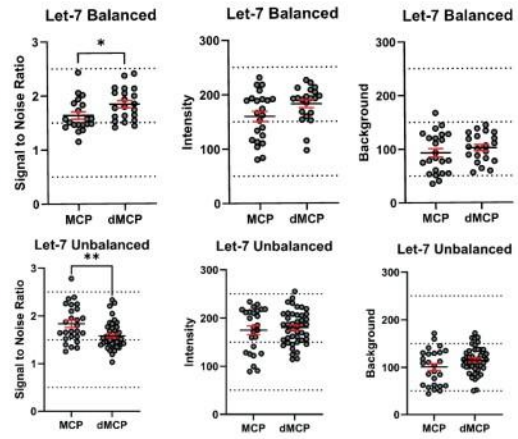


Figure 6. Effect of dMCP on fluorescence intensity and background in balanced and unbalanced let-7 worms. Graph created by student research using GraphPad Prism v9, 2025. Balanced statistics: t(38.83) = 2.177, p = 0.0357, d = 0.66 Unbalanced statistics: t(41.08) = 2.899, p = 0.0060, d = 0.72

Discussion

- **Let-7 detection:** Direct MS2-MCP visualization confirms that let-7 is actively transcribed during the L4 stage, providing the **first in vivo evidence** of its activation within the heterochronic pathway.
- **Let-7 vs. lin-4:** Let-7 exhibited a significantly higher SNR than lin-4 under identical conditions ($p < 0.001$) with a very large effect size (Cohen's d = 2.50), reflecting substantially greater transcriptional signal in let-7 relative to lin-4, despite similar raw fluorescence intensity. The comparable amplitudes are consistent with a **shared upstream regulatory mechanism**.
- **Balancing effect:** Balanced dMCP significantly improved SNR by reducing nuclear background ($p = 0.0022$), producing a large effect size (Cohen's d = 0.87). This supports the model that **stoichiometric balancing improves signal quality** by reducing background noise rather than increasing transcriptional output.
- **dMCP effect:** dMCP increased SNR only when balanced (Cohen's d = 0.66), but reduced SNR when unbalanced (Cohen's d = 0.72), demonstrating that **stoichiometric imbalance influences signal quality more than the dMCP construct.**

Conclusion and Future Research

- **Visualized endogenous let-7 transcription in vivo for the first time**
- Let-7 transcription occurs during the L4 stage
- Let-7 and lin-4 exhibited similar transcriptional amplitude
- Balanced MCP expression significantly improves MS2-MCP signal quality
- **Balanced dMCP provides the optimal environment for an ideal SNR**

- Perform a full developmental time course of let-7 transcription dynamics
- Test transcription factor mutants (lin-42, nhr-23, nhr-85, myrf-1) to identify upstream regulators
- Incorporate let-7 and lin-4 transcriptional data into timing and developmental models